KARTIK KARKERA

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Education

University of Texas, Dallas **May 2026**
Master of Science in Computer Science *GPA: 3.67 / 4.0*

Technical Skills

Programming Languages: Python, Java, SQL
Frontend: ReactJS, React Query, React Router
Backend: Flask, GraphQL, REST APIs
Cloud: AWS (Lambda, S3, EventBridge, Redshift), Apache Spark, Kafka
Databases: PostgreSQL, DynamoDB (Streams), Snowflake, Elasticsearch, MongoDB
Tools & DevOps: Git, GitHub Actions (CI/CD), Docker

Certifications

Databricks Certified Associate Developer for Apache Spark - Python Jan 2026
AWS Certified Solutions Architect – Associate (SAA-C03) Jul 2025
AWS Certified Data Engineer – Associate (DEA-C01) Sept 2025

Experience

PyCray Technologies Noida, India
Software Developer *Aug 2023 – Jun 2024*

- Architected serverless microservices processing algorithmic trading data, integrating DynamoDB Streams with AppSync to broadcast real-time position updates with sub 50ms latency.
- Developed a unified analytics engine aggregating trading metrics from hot (DynamoDB) and cold (PostgreSQL) storage, enabling strategic data-driven decisions and significantly better user engagement.
- Led code quality initiatives by establishing CI/CD workflows and conducting technical reviews, mentoring junior developers on cloud-native design patterns to reduce deployment friction.

PyCray Technologies Noida, India
Software Developer Intern *May 2023 – Aug 2023*

- Improved the carpooling app's ride search feature using Elasticsearch, making driver and rider matches more accurate and cutting search time by nearly half.
- Developed and optimized GraphQL APIs and integrated them with ReactJS-based dashboards to enable real-time user insights and reporting.
- Wrote Selenium scripts to automate Telegram group management, saving several weekly hours of manual work by handling login and scheduled messages automatically.

Trivia Softwares Mumbai, India
Software Developer Intern *Aug 2021 – Sept 2021*

- Developed a Django-based website with secure user authentication and email verification, improving sign-up security and creating a smoother onboarding experience.

Projects

Traceline - Pipeline Anomaly Detection | *Python, Flask, Pandas, NumPy* **1st Place - TidalHack @ TAMU**

- Built an intelligent system for RCP Inc. that aligns 15+ years of inspection data to track corrosion growth and predict failure risks.
- Engineered a relative-matching algorithm using fixed hardware anchors to overcome "odometer drift".
- Contributed to a React-based 3D visualization dashboard to help users track anomaly evolution over time.

Data-Centric AI Challenge | *Python, 3LC, AWS* **3rd Place - 3LC x AWS Hackathon**

- Optimized a ResNet-18 classifier under strict rules (no model/architecture changes, no external data or pretrained weights); pushed accuracy purely via data quality improvements to 94% validation accuracy
- Used 3LC Dashboard to iteratively relabel low-confidence samples, rebalance classes, and adjust per-sample weights from embeddings, demonstrating a fully data-centric approach.